

P1. Let $J_m(\lambda) = \begin{pmatrix} \lambda & 1 & & 0 \\ & \lambda & 1 & \\ & & \ddots & \ddots \\ 0 & & & \lambda \end{pmatrix} = \lambda I + \begin{pmatrix} 0 & 1 & & \\ & 0 & 1 & \\ & & \ddots & \ddots \\ & & & 0 \end{pmatrix}$

Let $A_n = \begin{pmatrix} 0 & 1 & & \\ & 0 & 1 & \\ & & \ddots & \ddots \\ & & & 0 \end{pmatrix}^n$ $A_1 = \begin{pmatrix} 0 & 1 & & \\ & 0 & 1 & \\ & & \ddots & \ddots \\ & & & 0 \end{pmatrix}$

Use Mathematical induction to prove.

$$A_n = \begin{pmatrix} \overbrace{0 \dots 0}^{n's} & 1 & & 0 \\ & 0 & 1 & \\ & & \ddots & \ddots \\ 0 & & & \underbrace{0 \dots 0}_{n's} \end{pmatrix}$$

case $n=1$, $A_1 = \begin{pmatrix} 0 & 1 & & \\ & 0 & 1 & \\ & & \ddots & \ddots \\ & & & 0 \end{pmatrix}$ ✓

if case n ✓, then case $n+1$.

$$A_{n+1} = A_1 \cdot A_n = \begin{pmatrix} 0 & 1 & & \\ & 0 & 1 & \\ & & \ddots & \ddots \\ & & & 0 \end{pmatrix} \begin{pmatrix} \overbrace{0 \dots 0}^{n's} & 1 & & 0 \\ & 0 & 1 & \\ & & \ddots & \ddots \\ 0 & & & \underbrace{0 \dots 0}_{n's} \end{pmatrix} = (a_{ij})_{m \times m}^{(n+1)}$$

consider $a_{i,i+n+2}^{(n+1)} = \sum_{k=1}^m a_{i,k}^{(1)} a_{k,i+n+2}^{(n)}$ ($j = i+n+2$).

only when $k=i+1$, $a_{i,i+1} = a_{i+1,i+n+2} = 0$.

otherwise $a_{i,k} = a_{k,i+n+2} = 0$.

for any $j \neq i+n+2$, $a_{i,j}^{(n+1)} = \sum_{k=1}^m a_{i,k}^{(1)} a_{k,j}^{(n)}$.

for any k , $a_{i,k}^{(1)} a_{k,j}^{(n)} = 0$ so $a_{i,j}^{(n+1)} = 0$.

so $A_{n+1} = \begin{pmatrix} \overbrace{0 \dots 0}^{n+1's} & 1 & & \\ & 0 & 1 & \\ & & \ddots & \ddots \\ & & & 0 \end{pmatrix}$ when $n \geq m$, $A_n = 0$

by mathematical induction.

$$A_n = \begin{pmatrix} \overbrace{0 \dots 0}^{n's} & 1 & & 0 \\ & 0 & 1 & \\ & & \ddots & \ddots \\ 0 & & & \underbrace{0 \dots 0}_{n's} \end{pmatrix} \quad | \leq n < m$$

when $n \geq m$ $A_n = 0$.

$$\begin{aligned}
 \text{so } J_m(\lambda)^n &= (\lambda I + A_1)^n = \sum_{k=0}^n \binom{n}{k} (\lambda I)^k \cdot A_1^{n-k} \\
 &= \sum_{k=0}^{n-1} \binom{n}{k} \lambda^k A_{n-k} + \underline{\underline{\lambda^n I}}
 \end{aligned}$$

$$A_{n-k} = \begin{pmatrix} \overset{n-k}{\overbrace{0 \dots 0}} & 1 & 0 & \dots & 0 \\ & & \ddots & & \\ 0 & & & 1 & \\ & & & & \ddots \\ & & 0 & & & 0 \end{pmatrix} = a_{ij}^{(n-k)} = \begin{cases} 1 & j = i + n - k \Rightarrow k = n - j + i \\ 0 & j \neq i + n - k \end{cases}$$

$$\begin{aligned}
 \text{so } J_m(\lambda)^n &= (c_{ij}) = \begin{cases} \binom{n}{n-j+i} \lambda^{n-j+i}, & 0 \leq j-i \leq \min\{n, m-1\} \\ 0, & \text{otherwise} \end{cases} \quad (\bar{j}=i, c_{ii} = \lambda^n) \\
 &= \begin{cases} \binom{n}{j-i} \lambda^{n-j+i}, & 0 \leq j-i \leq \min\{n, m-1\} \\ 0, & \text{otherwise} \end{cases}
 \end{aligned}$$

$$= \begin{pmatrix} \lambda^n \binom{n}{1} \lambda^{n-1} \binom{n}{2} \lambda^{n-2} \dots \binom{n}{m-1} \lambda^{n-m+1} & & & \\ \lambda^n \binom{n}{1} \lambda^{n-1} \dots \binom{n}{m-2} \lambda^{n-m+2} & & & \\ & \ddots & & \\ & & \binom{n}{1} \lambda^{n-1} & \\ & & & \lambda^n \end{pmatrix} \quad \binom{n}{k} = 0, \text{ if } k > n$$

P2. $A_n = (a_{ij})_{n \times n} = \begin{cases} 2 & i=j \\ (-1)^{|i-j|} & i \neq j \end{cases} = \begin{pmatrix} 2 & -1 & 1 & -1 & \dots & (-1)^{n-1} \\ -1 & 2 & -1 & & & \\ 1 & -1 & 2 & & & \\ \vdots & & & \ddots & & \\ (-1)^{n-1} & & & & 2 \end{pmatrix}$

for any k -th column, add the number of $(k-1)$ th column.

from $k=n, n-1, \dots, 2$ one by one.

then $|A_n| = \begin{vmatrix} 2 & -1 & 1 & \dots & (-1)^{n-1} \\ -1 & 2 & -1 & & \\ \vdots & -1 & 2 & & \\ \vdots & & & \ddots & \\ (-1)^{n-1} & & & & 2 \end{vmatrix} = \begin{vmatrix} 2 & -1 & 1 & \dots & (-1)^{n-1} & 0 & 0 \\ -1 & 2 & -1 & & & \vdots & 0 \\ \vdots & -1 & 2 & & & \vdots & 0 \\ \vdots & & & \ddots & & 2 & 0 \\ (-1)^{n-1} & & & & & -1 & 0 \end{vmatrix}$

$= \dots = \begin{vmatrix} 2 & 1 & 0 & \dots & 0 \\ -1 & 1 & 1 & & 0 \\ \vdots & 0 & 1 & & 0 \\ -1 & 0 & & \ddots & 0 \\ (-1)^{n-1} & 0 & & & 0 \end{vmatrix} = 2 \begin{vmatrix} 1 & 1 & & & \\ & 1 & & & \\ & & \ddots & & \\ & & & 1 & \\ & & & & 1 \end{vmatrix}$
↓
first column

$+ \begin{vmatrix} 1 & 1 & & & \\ & 1 & & & \\ & & \ddots & & \\ & & & 1 & \\ & & & & 1 \end{vmatrix} + \begin{vmatrix} 1 & 0 & & & \\ 0 & 1 & & & \\ & 0 & 1 & & \\ & & 0 & \ddots & \\ & & & 0 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 0 & & & \\ 0 & 1 & & & \\ & 0 & 1 & & \\ & & 0 & \ddots & \\ & & & 0 & 1 \end{vmatrix} + \dots + \begin{vmatrix} 1 & & & & \\ & 1 & & & \\ & & \ddots & & \\ & & & 1 & \\ & & & & 1 \end{vmatrix}$

because $\begin{vmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{vmatrix} = \begin{vmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{vmatrix} = 1$

and if $M = \begin{pmatrix} P_{a \times a} & 0 \\ 0 & Q_{b \times b} \end{pmatrix}$, then $|M| = |P_{a \times a}| |Q_{b \times b}|$.

so $A_n = 2 + \underbrace{1 + 1 + \dots + 1}_{(n-1)'s} = n+1$.